

Customer Churn Analysis

Task 5 – Executive Analytics Capstone

Executive objective: Turn the 15-customer churn analysis into decision-ready insights, quantify the observed impact, state limitations, and recommend practical retention actions.

1. Executive Snapshot

KPI	Result
Customers	15
Churned customers	7
Churn rate	46.67%
Retention rate	53.33%
Observed total customer value (TotalCharge)	■31,327.18
Value associated with churned customers	■3,109.51
Share of observed value associated with churned customers	9.93%

The sample contains 15 customers, of whom 7 churned, producing an observed churn rate of 46.67%. The churned group represents ■3,109.51 of the ■31,327.18 total customer value shown in the dataset. Because this is a small synthetic/sample dataset, these figures should be treated as directional rather than forecasts.

2. Five Business Insights

1. Churn is material in the sample: 7 of 15 customers churned (46.67%).
2. Month-to-month contracts are the highest-risk contract type: 7 of 7 churned (100.00%).
3. Churn is concentrated among shorter-tenure customers: the churned group averages 7.0 months of tenure versus 29.1 months for retained customers.
4. Churned customers have more support tickets on average (4.3) than retained customers (1.0), suggesting support friction is a useful retention signal.
5. The observed value attached to churned customers is ■3,109.51, or 9.93% of the sample's TotalCharge; retention should therefore prioritize high-value as well as high-risk accounts.

3. Segment and Contract View

Contract	Customers	Churned	Churn Rate
Month-to-month	7	7	100.00%
One Year	4	0	0.00%
Two Year	4	0	0.00%

Subscription mix:

- Basic: 7 customers, 5 churned (71.43%); observed TotalCharge ■3,349.33.
- Enterprise: 4 customers, 0 churned (0.00%); observed TotalCharge ■23,098.46.
- Pro: 4 customers, 2 churned (50.00%); observed TotalCharge ■4,879.39.

4. Recommended Actions

Action 1 – Retention outreach: Prioritize high-risk customers, especially month-to-month customers with high support-ticket counts and very short tenure.

Action 2 – Contract conversion: Offer suitable incentives or value messaging to move month-to-month customers toward longer commitments.

Action 3 – Support intervention: Review recurring support issues for customers with 4+ tickets and close unresolved issues quickly.

Action 4 – Value-based prioritization: Combine churn risk with TotalCharge so retention resources focus on customers where both risk and business value are high.

Action 5 – Monitoring: Repeat the analysis on a larger, real historical dataset and track churn by cohort, contract, subscription, tenure, and support burden.

5. Risk Prioritization Model

The simple rule-based model used for the capstone assigns points for month-to-month contracts (+3), 4 or more support tickets (+2), and tenure below 6 months (+2). Scores 5–7 are treated as High risk, 3–4 as Medium, and 0–2 as Low. This model is intentionally transparent and should not be interpreted as a trained predictive model.

Customer	Contract	Tenure	Support	Risk	Recommendation
CUST-1010	Month-to-month	4	4	High (7)	Immediate retention outreach; offer contract upgrade
CUST-1007	Month-to-month	3	6	High (7)	Immediate retention outreach; offer contract upgrade
CUST-1014	Month-to-month	2	5	High (7)	Immediate retention outreach; offer contract upgrade
CUST-1005	Month-to-month	8	4	High (5)	Immediate retention outreach; offer contract upgrade
CUST-1003	Month-to-month	6	5	High (5)	Immediate retention outreach; offer contract upgrade
CUST-1012	Month-to-month	14	3	Medium (3)	Proactive engagement and contract review
CUST-1001	Month-to-month	12	3	Medium (3)	Proactive engagement and contract review
CUST-1008	Two Year	48	1	Low (0)	Maintain relationship and monitor
CUST-1015	Two Year	40	1	Low (0)	Maintain relationship and monitor
CUST-1004	Two Year	36	0	Low (0)	Maintain relationship and monitor
CUST-1011	Two Year	30	0	Low (0)	Maintain relationship and monitor
CUST-1002	One Year	24	1	Low (0)	Maintain relationship and monitor
CUST-1009	One Year	15	2	Low (0)	Maintain relationship and monitor
CUST-1013	One Year	22	1	Low (0)	Maintain relationship and monitor
CUST-1006	One Year	18	2	Low (0)	Maintain relationship and monitor

6. Assumptions and Limitations

- The dataset contains only 15 customers, so percentages can change substantially with one or two records.
- The dataset appears to be a synthetic/sample dataset; it should not be used to claim real-world causal relationships.

- TotalCharge is treated as the observed customer-value field; it is not the same as future revenue or guaranteed revenue at risk.
- Support tickets are treated as an association signal, not proof that support problems cause churn.
- The rule-based risk score is explainable and reproducible, but it is not statistically validated or a machine-learning probability.

7. Executive Conclusion

The immediate business priority is to protect customers who combine short tenure, month-to-month commitment, and higher support burden. A practical retention program should start with targeted outreach, support resolution, and contract-conversion offers, while using customer value to prioritize effort. The next analytical step is to validate these signals on a much larger historical dataset and measure whether interventions reduce churn.